



AI CAREER COMPANION AGENT FOR INTERNSHIP MATCHING AND INTERVIEW PREPARATION

PROJECT REPORT

Project Title: AI Career Companion Agent for Internship Matching and Interview Preparation

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1. Project Overview

1.1 Project Name

AI Career Companion Agent for Internship Matching and Interview Preparation

1.2 What is the Project?

The **AI Career Companion Agent for Internship Matching and Interview Preparation** is an AI-powered career assistance platform designed to help students and job seekers find suitable internship opportunities and prepare for their career journey. The system analyzes a user's resume to extract relevant information such as skills, education, and experience, and uses this information to identify suitable internship opportunities through semantic matching.

The platform also provides additional career-support features such as **resume analysis, internship matching, skill-gap analysis, personalized cover letter generation, and interview preparation**. By combining resume processing, AI-based analysis, and semantic search, the system provides users with personalized assistance throughout the internship application process.

1.3 Problem Statement

Students and early-career candidates often face difficulties in finding internships that match their skills, qualifications, and career interests. Manually searching through numerous opportunities can be time-consuming, while assessing whether a candidate meets specific internship requirements can be challenging.

Additionally, candidates may struggle to identify skill gaps, prepare for interviews, and create customized cover letters. Existing job-search platforms often provide general listings without offering a complete, personalized career-support workflow.

The proposed system addresses these challenges using **AI-based resume analysis and semantic internship matching** to identify relevant opportunities and provide personalized application and interview support.

1.4 Target Users

The primary target users of the application are:

- **Students** looking for internships and career opportunities.
- **Undergraduate and graduate students** seeking opportunities relevant to their academic background and skills.
- **Freshers and early-career candidates** preparing for internship or entry-level applications.
- **Job seekers** who want personalized assistance with resumes, cover letters, and interview preparation.

2. Project Objective

The primary objective of the **AI Career Companion Agent for Internship Matching and Interview Preparation** is to develop an AI-powered platform that helps students and early-career candidates discover suitable internship opportunities and prepare effectively for the application process.

Objectives:

- To analyze resumes and extract relevant information such as skills, education, and experience.
- To match candidates with suitable internship opportunities using semantic search and AI-based matching techniques.
- To identify skill and requirement gaps between a candidate's profile and internship opportunities.
- To generate personalized cover letters based on the candidate's profile and selected internship.
- To provide AI-powered interview preparation and career guidance.
- To reduce the time and effort required to search for relevant internships and prepare applications.
- To provide a unified platform for resume analysis, internship discovery, application support, and interview preparation.

3. Technology Stack

Category	Technology
Programming Languages	Python, JavaScript (ES6+), HTML5, CSS3
Backend Framework	FastAPI
Frontend	Vanilla JavaScript, HTML5, CSS3
AI / LLM	Groq API
Vector Search / RAG	FAISS, Vector RAG
Data Extraction / Parsing	PyMuPDF, python-docx, Regular Expressions
Database	SQLite
Local Storage	Browser localStorage
API / Documentation	REST API, Swagger / OpenAPI
Testing	Pytest
Version Control	Git, GitHub

4. Project Requirements

Functional Requirements

- User should be able to register and log in.
- User should be able to upload a resume.
- System should extract relevant information from the resume.
- System should analyse the resume using AI.
- User should be able to find suitable internship opportunities.
- System should provide internship matches based on the user's profile and skills.
- System should provide ATS/resume analysis and skill-gap information.
- User should be able to generate personalized cover letters.
- User should be able to access interview preparation and AI-based career assistance.

Non-Functional Requirements

- Good response time
- Secure user authentication
- User-friendly interface
- Reliable AI responses
- Proper error handling
- Maintainable and scalable system

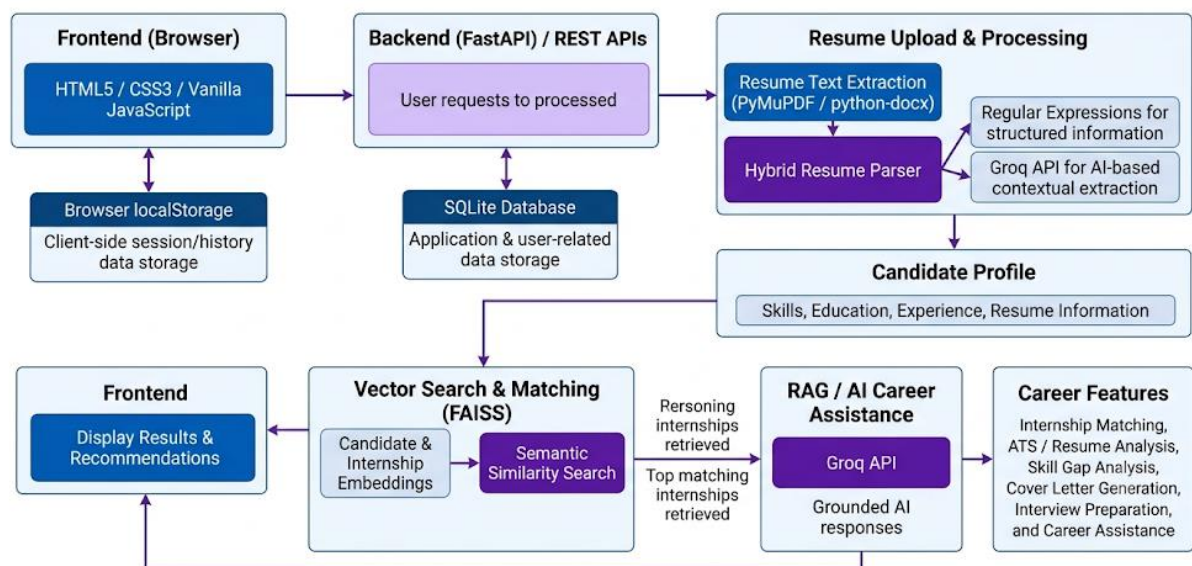
5. System Architecture and Workflow

5.1 System Architecture

The system follows a **client-server architecture** where the Vanilla JavaScript frontend communicates with the FastAPI backend through REST APIs. The backend processes resumes, performs AI-based analysis using the Groq API, and uses FAISS-based vector search for internship matching.

Architecture Diagram:

System Architecture / Workflow

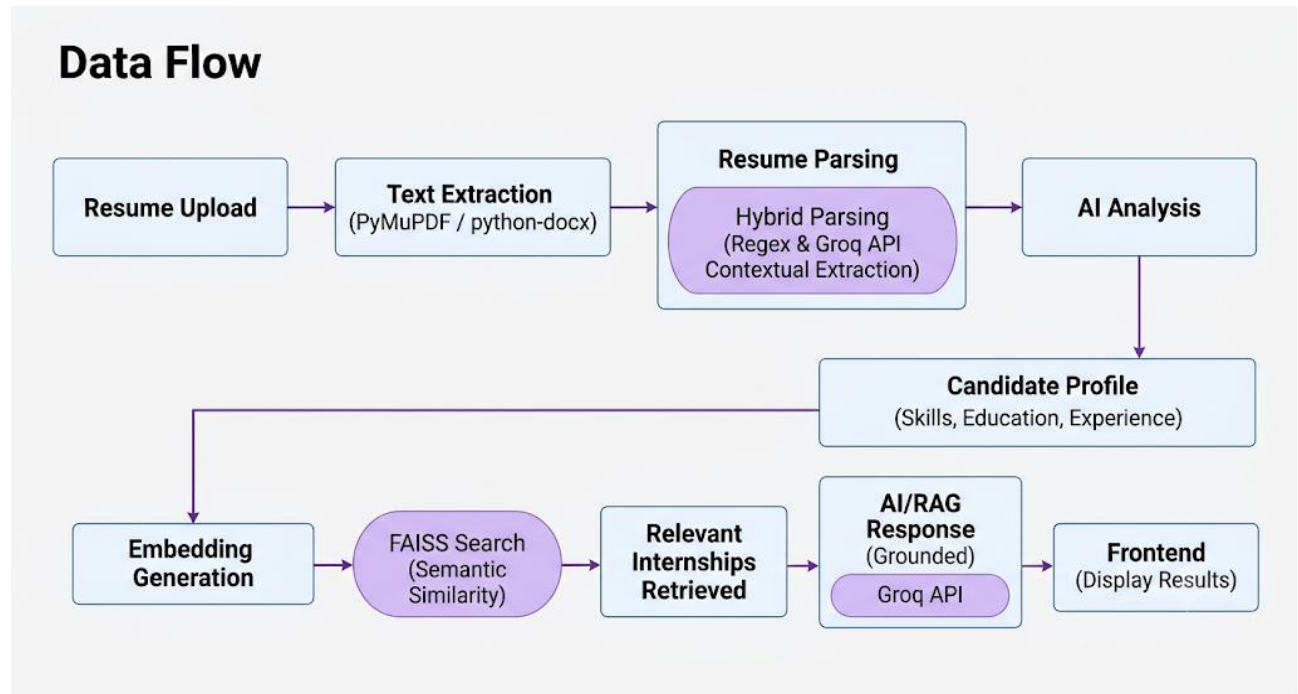


5.2 Main Components

- **Frontend:** Vanilla JavaScript, HTML, and CSS interface.
- **FastAPI Backend:** Handles API requests and application logic.
- **Authentication:** Manages user registration, login, sessions, and account operations.
- **Resume Parser:** Extracts information from PDF/DOCX resumes using text extraction, regular expressions, and AI.
- **Groq API:** Performs AI-based resume analysis and generates career-related responses.
- **FAISS Vector Database:** Stores and searches internship embeddings for semantic matching.
- **SQLite Database:** Stores application and user-related data.
- **RAG Module:** Retrieves relevant internship information and provides grounded AI responses.

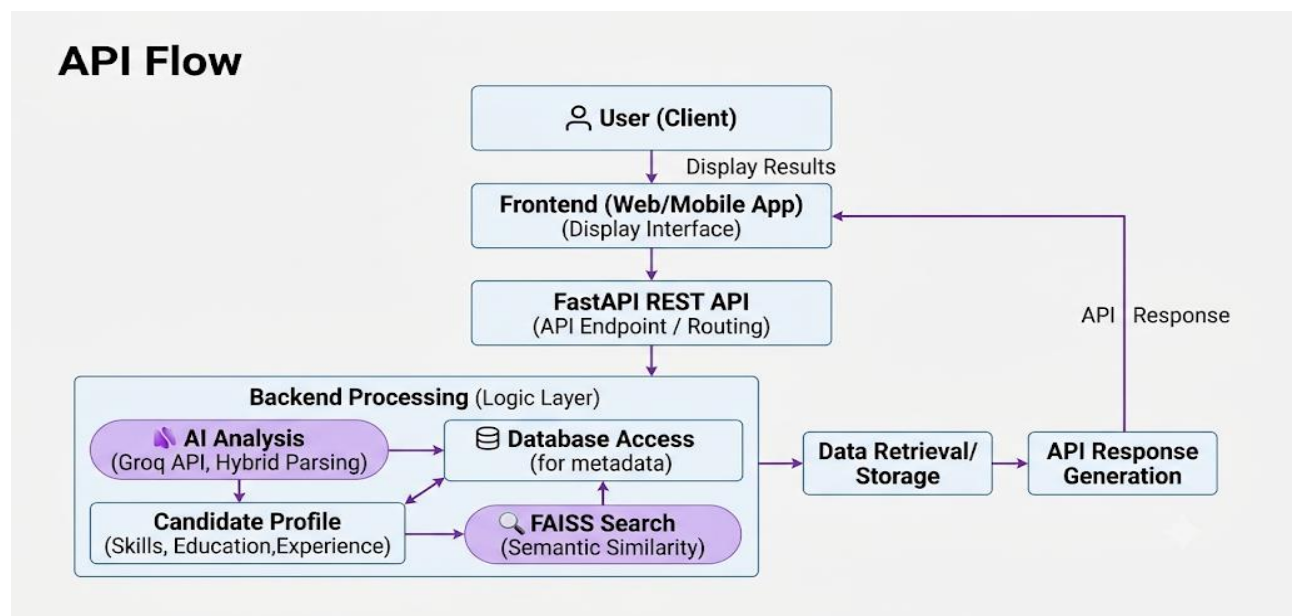
5.3 Data Flow

The system extracts and analyzes the resume using Regex and the Groq API to create a candidate profile. The profile is then matched using FAISS and RAG to generate relevant results, as shown in the diagram below.



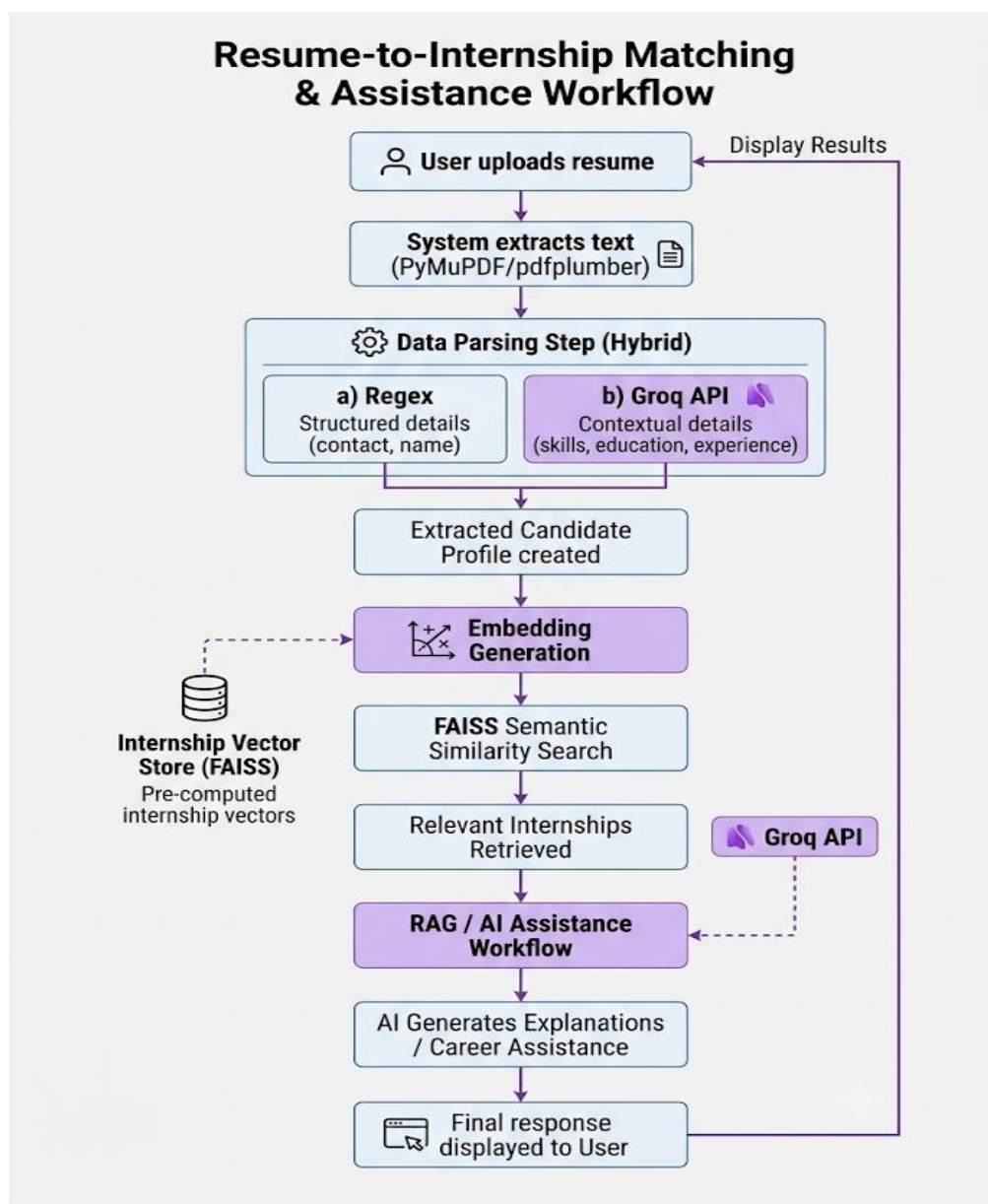
5.4 API Flow

The frontend sends requests to the FastAPI REST API for processing, and the backend returns the processed results to the frontend, as shown in the diagram below.



5.5 AI / RAG Workflow

1. User uploads a resume.
2. The system extracts text from the resume.
3. Regex is used for structured information such as contact details.
4. The Groq API analyzes contextual information such as skills, education, and experience.
5. The extracted candidate profile is converted into an embedding representation.
6. Internship information is stored as vectors in FAISS.
7. FAISS performs semantic similarity search to find relevant internships.
8. Retrieved internship information is used by the RAG workflow.
9. The AI generates relevant explanations or career assistance.
10. The final response is displayed to the user.



6. Product Backlog

ID	Feature / Task	Priority	Status
P001	Project Initialization & Repository Setup	High	Done
P002	User Authentication & Guest Session Handling	High	Done
P003	Resume & Document Upload Pipeline (.PDF / .DOCX)	High	Done
P004	Hybrid Resume Parsing (LLM + Regex Fallback)	High	Done
P005	Vector Embedding Generation & FAISS Semantic Matching	High	Done
P006	Grounded RAG Explanation & Fit Scoring	High	Done
P007	Automated ATS Resume Optimization Scorer & Audit	High	Done
P008	Dynamic Skill Gap Analysis & Learning Recommendations	High	Done
P009	AI-Powered Custom Cover Letter Generator	Medium	Done
P010	Floating Career Copilot Chat Interface	Medium	Done
P011	Multi-Session Interview Prep & Document Q&A Agent	High	Done
P012	Responsive Collapsible Sidebar & SaaS Navigation UI	Medium	Done
P013	Dark / Light Theme Mode Toggle	Low	Done
P014	Interactive Animated Landing Page Canvas	Low	Done
P015	Comprehensive System & End-to-End Integration Testing	High	Done

7. Sprint Planning

The project development was divided into smaller development cycles called **Sprints**. Each sprint focused on a specific development goal and helped in completing the project systematically.

Sprint 1

Duration: 1 Week

Goal: Project Setup

Tasks:

- Understand project requirements
 - Create project structure
 - Set up GitHub repository
 - Set up database
 - Set up development environment
-

Sprint 2

Duration: 1 Week

Goal: Backend Development

Tasks:

- Create APIs
 - Implement authentication
 - Create database models
 - Implement document upload
-

Sprint 3

Duration: 1 Week

Goal: AI/RAG Implementation

Tasks:

- Implement document processing
- Implement text processing
- Implement embedding generation

- Set up vector database
 - Integrate LLM
 - Implement RAG pipeline
-

Sprint 4

Duration: 1 Week

Goal: Final Integration & Testing

Tasks:

- Integrate frontend with backend
- Perform API testing
- Perform feature testing
- Fix bugs
- Prepare final project demo

8. Sprint Tasks & Status

Sprint	Task	Status
Sprint 1	Requirement Analysis & Scope Definition	Done
Sprint 1	Project Directory & Repository Setup	Done
Sprint 1	Environment Configuration & Dependency Setup	Done
Sprint 1	Local State Architecture & Storage Schema	Done
Sprint 2	FastAPI Core Endpoints Setup	Done
Sprint 2	Authentication & Guest Session Handling	Done
Sprint 2	Document Intake & Multi-Format Parsing (.PDF / .DOCX)	Done
Sprint 2	Hybrid Parsing Engine (LLM + Regex Fallbacks)	Done
Sprint 3	Vector Embeddings & FAISS Index Creation	Done
Sprint 3	Semantic Matching & Grounded RAG Explanations	Done
Sprint 3	ATS Scoring Engine & Gap Analysis Pipeline	Done
Sprint 3	Multi-Session Interview Prep & Document Q&A Agents	Done
Sprint 4	Dashboard View & API Integration	Done
Sprint 4	Collapsible Sidebar & Theme Mode Switching	Done
Sprint 4	Career Copilot & Interactive Landing UI	Done
Sprint 4	End-to-End API & User Workflow Testing	Testing
Sprint 4	Bug Fixing & Demo Packaging	Done

Kanban Board View

To Do	In Progress	Testing	Done
—	—	End-to-End Parsing & RAG Testing	Project & Repository Setup
		Local Server Deployment	Authentication & Sessions
		Multi-Device Responsive UI Testing	Hybrid Resume Parser
			FAISS Semantic Matching
			ATS Audit & Gap Analysis
			Cover Letter Generator
			Interview Prep Agent
			Career Copilot
			Interactive Landing Page Canvas
			Collapsible Sidebar & Theme Switcher
			Dashboard & API Integration
			Application Tracker
			Document Q&A
			AI Guardrails & Policy RAG

9. Implementation and Key Features

The project was implemented as an AI-powered career assistance platform that combines resume analysis, internship matching, application tracking, career assistance, and interview preparation.

Feature 1: Landing Page

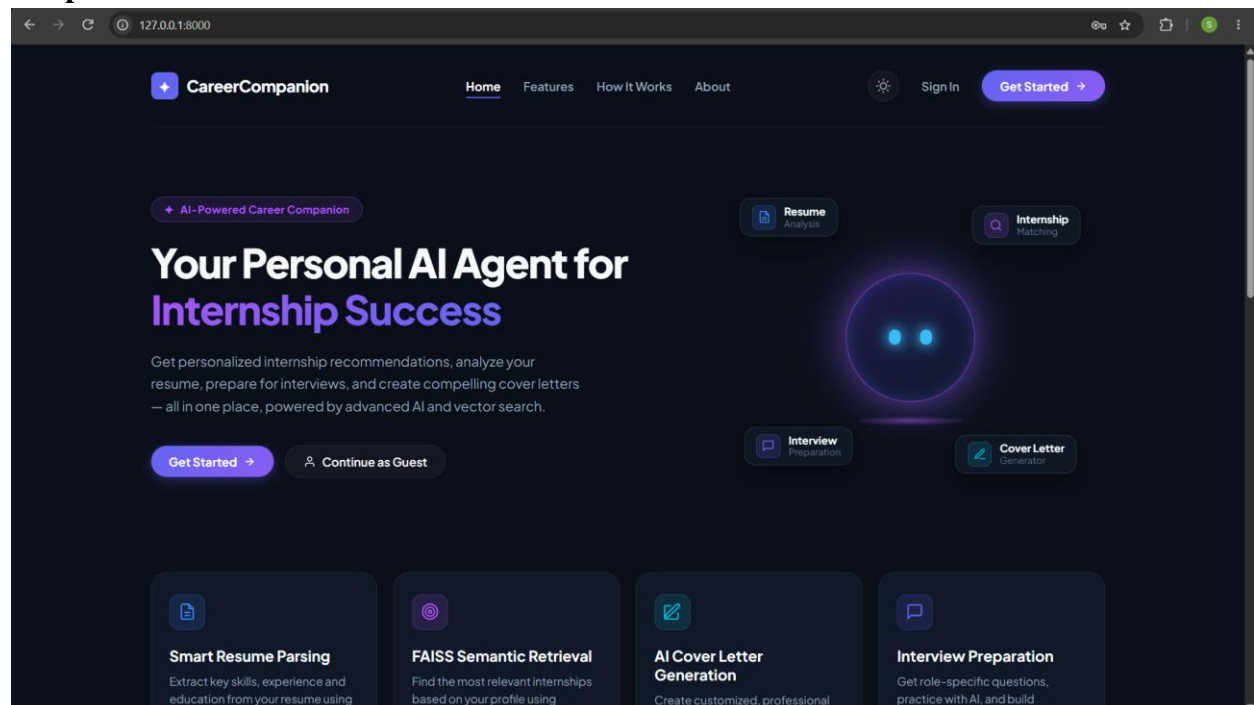
Description:

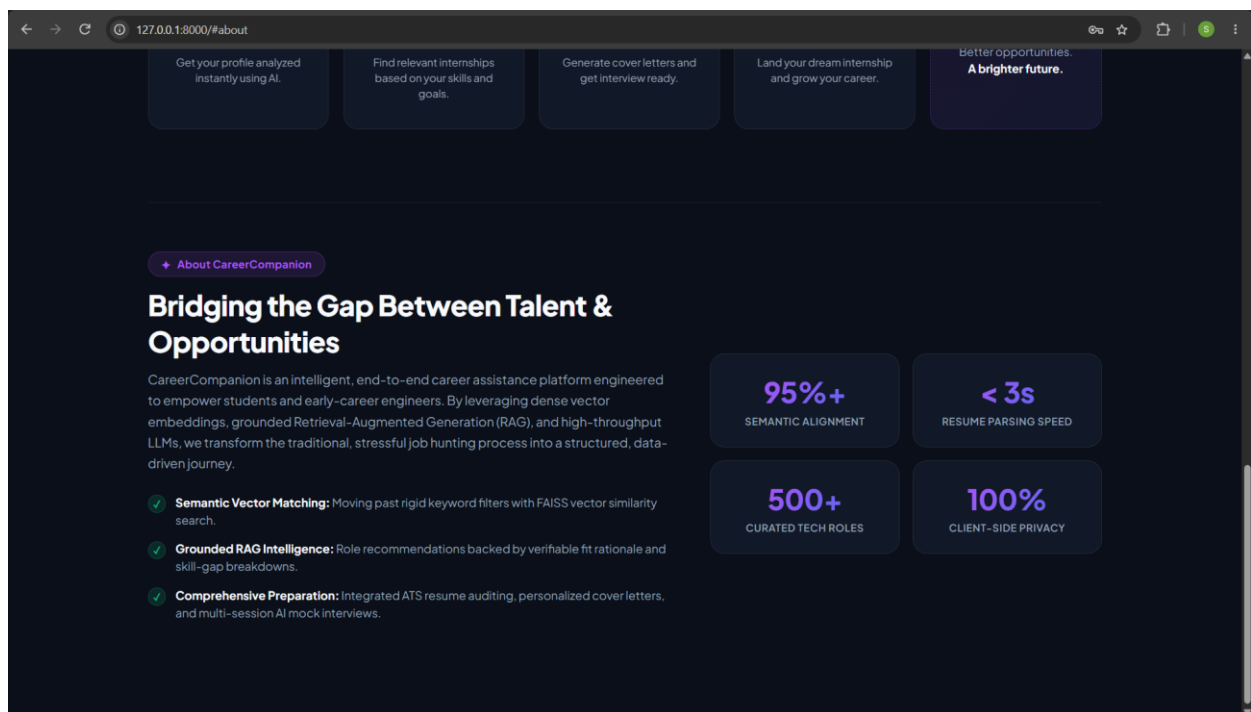
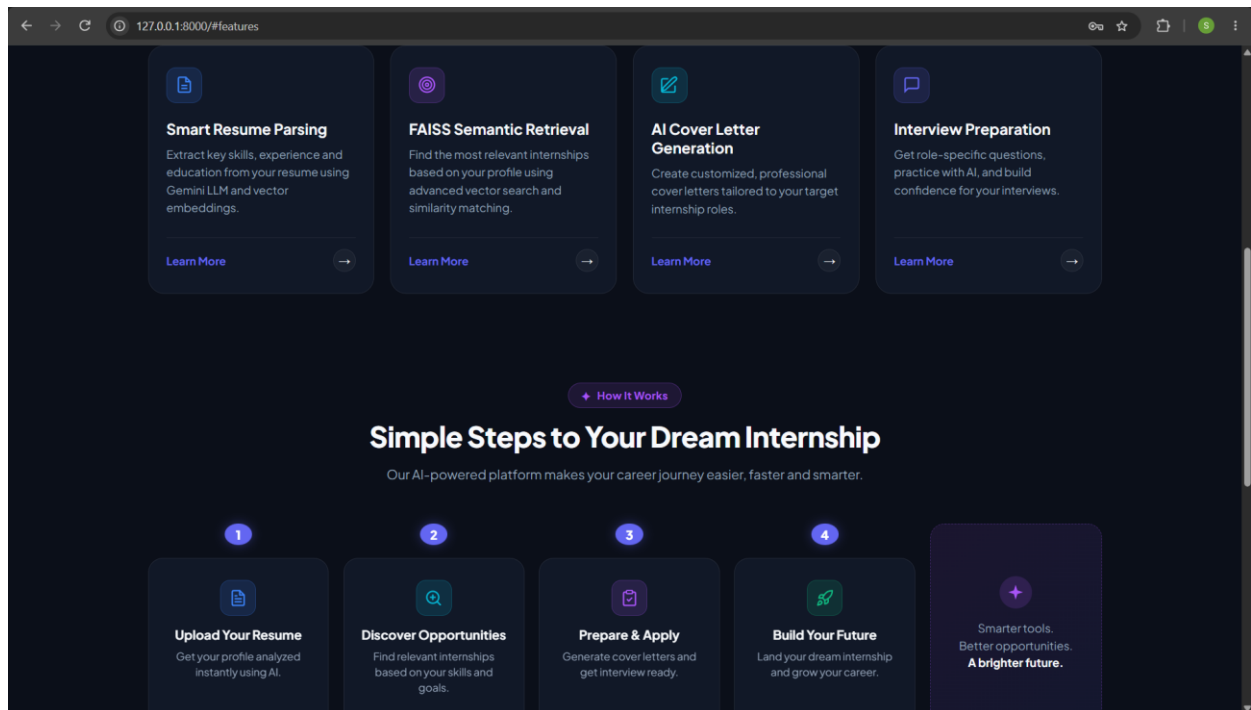
The platform provides an interactive landing page that introduces the application and its major career assistance features.

Implementation:

An animated canvas interface was integrated into the landing page to provide an interactive and engaging user experience.

Output:





Feature 2: User Authentication & Session

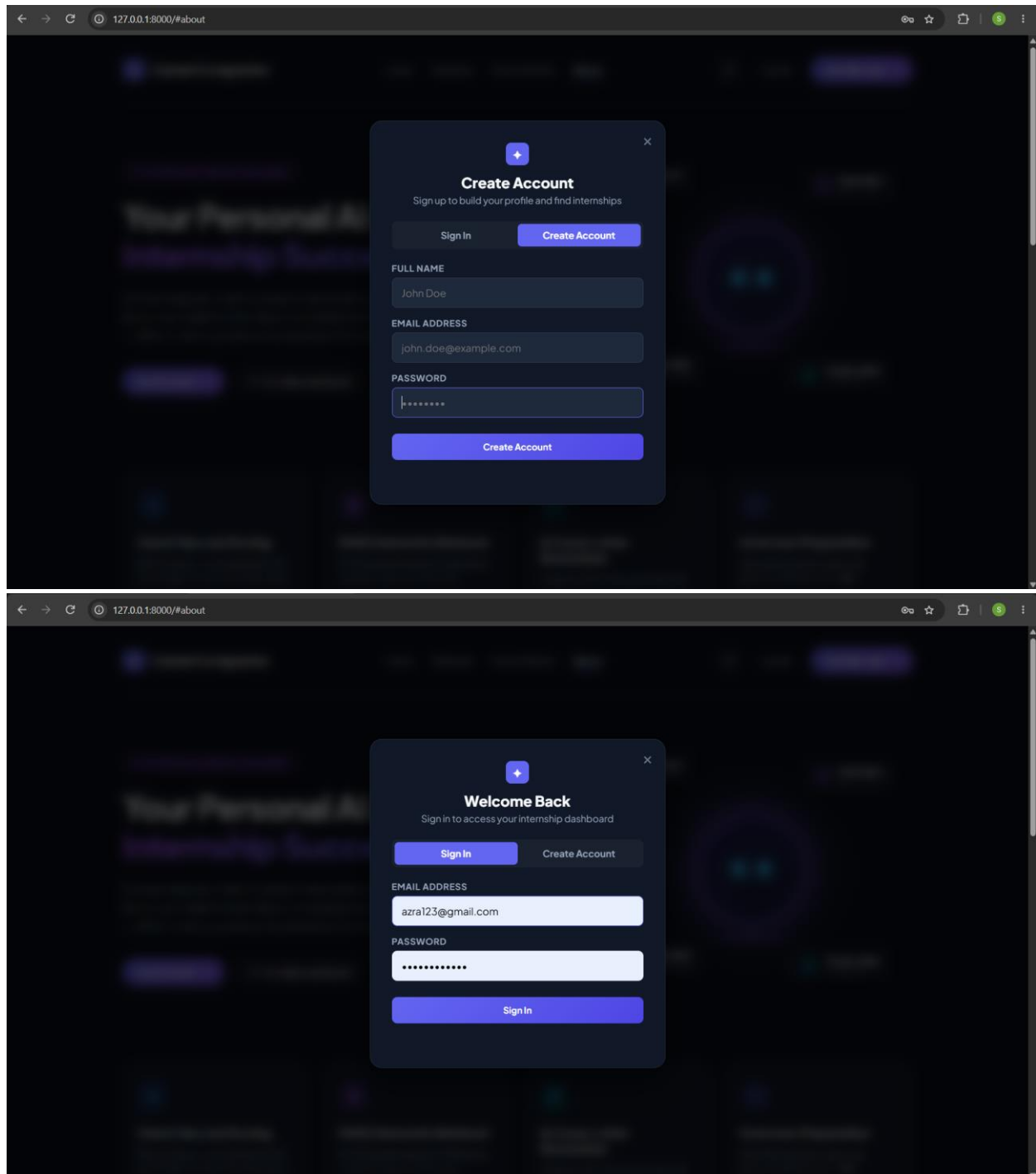
Description:

Users can register, log in, manage their sessions, and access personalized career features.

Implementation:

Authentication is implemented using FastAPI APIs, with browser localStorage used for client-side session and application state.

Output:



Feature 3: Resume Upload & Parsing Using Hybrid Resume Parsing & AI Analysis

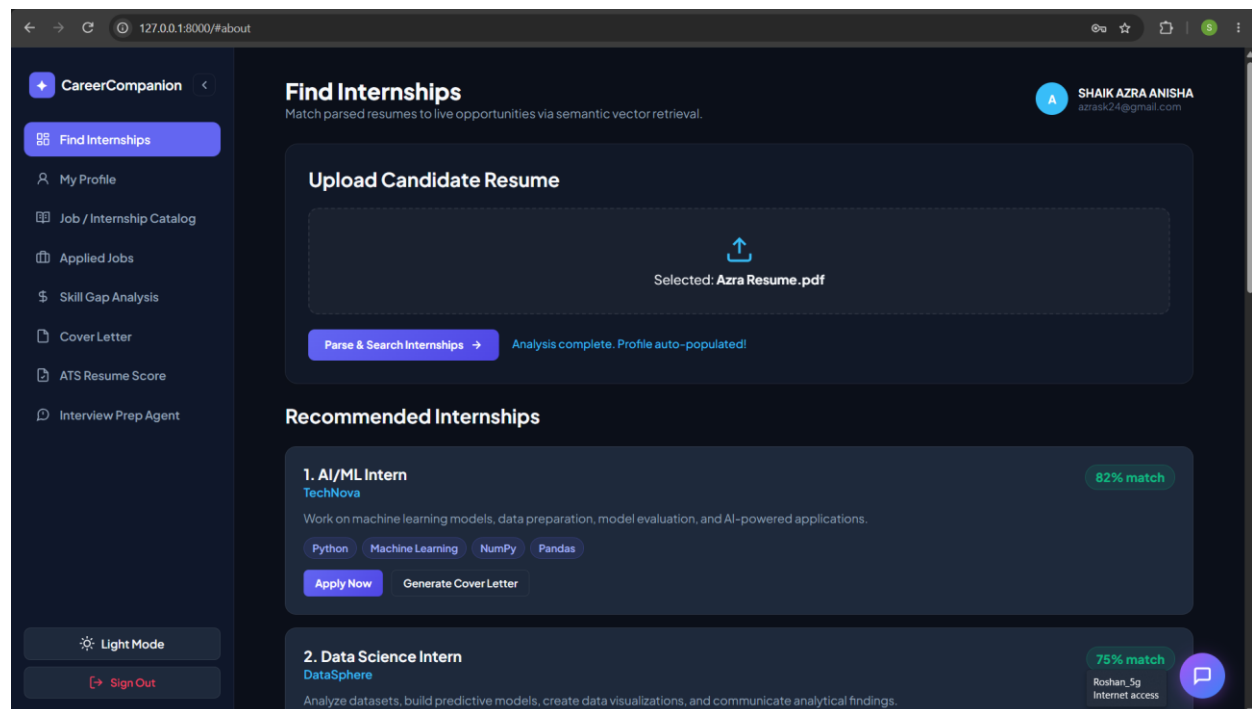
Description:

Users can upload resumes in PDF or DOCX format for automated extraction and AI-based analysis.

Implementation:

The system extracts resume content using PyMuPDF and python-docx, then combines Regex-based extraction with the Groq API to generate a structured candidate profile containing skills, education, experience, and other relevant details.

Output:



Feature 4: Internship Matching

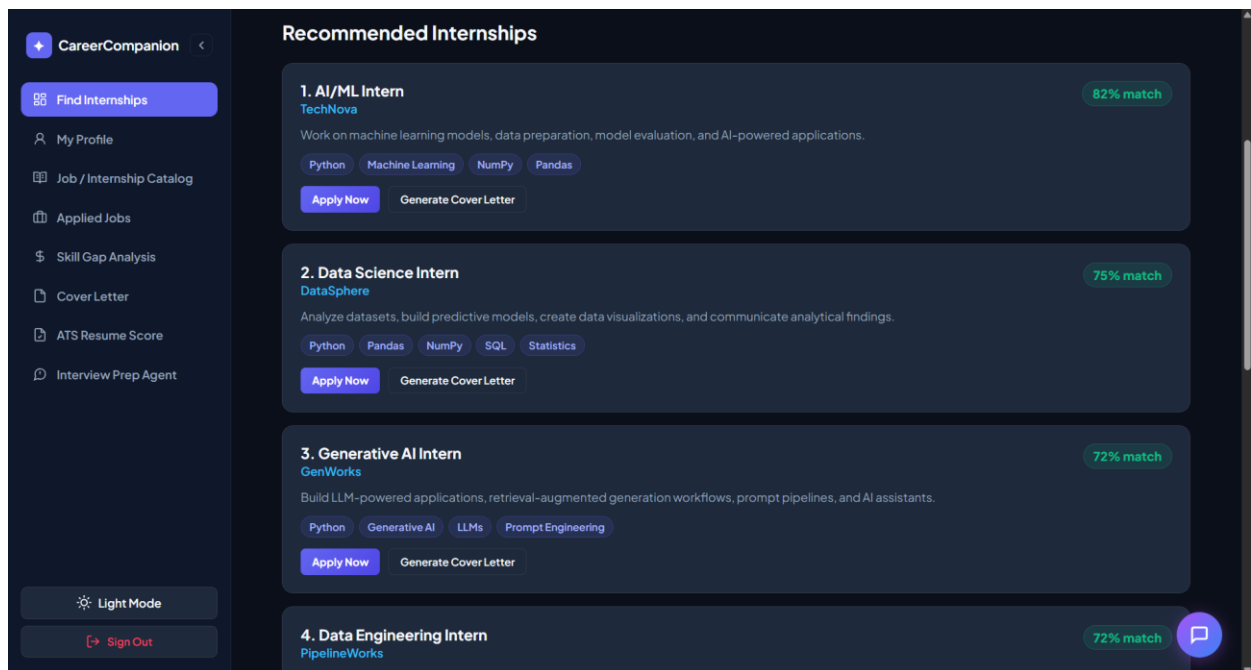
Description:

The system recommends internships that match the candidate's profile, skills, and qualifications.

Implementation:

Candidate and internship information are converted into vector embeddings, and FAISS performs semantic similarity search to identify relevant opportunities.

Output:



Feature 5: RAG-Based Internship Explanation

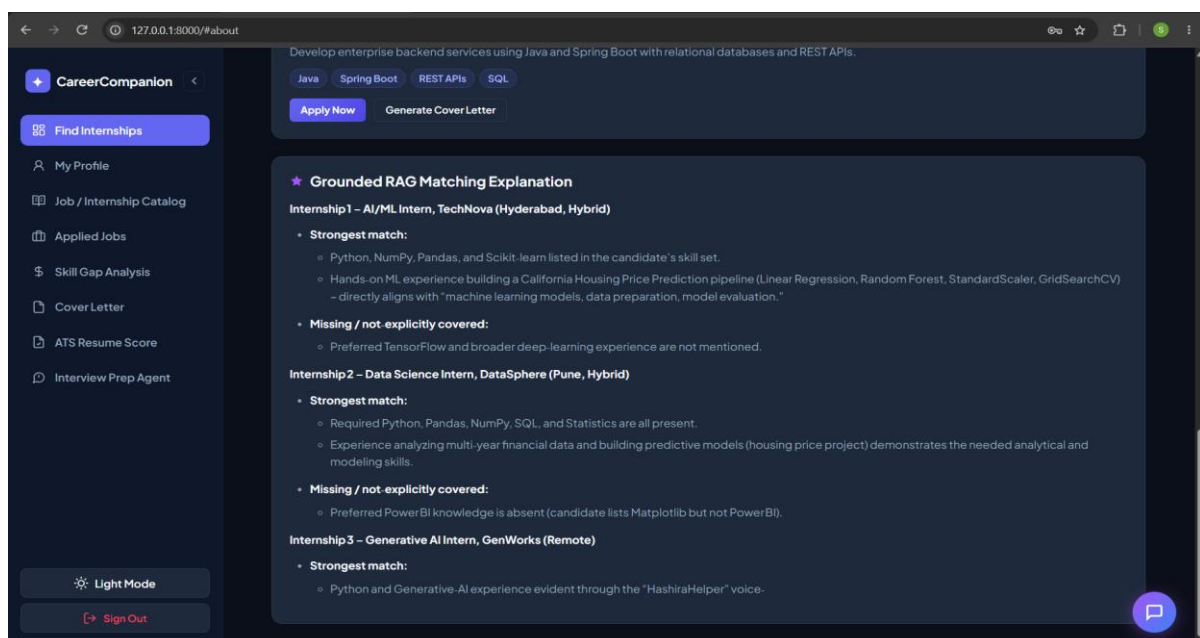
Description:

The system explains why a particular internship is relevant to the candidate.

Implementation:

Relevant internship information is retrieved through vector search and provided as context to the Groq API to generate grounded explanations.

Output:



Feature 6: User Profile Section

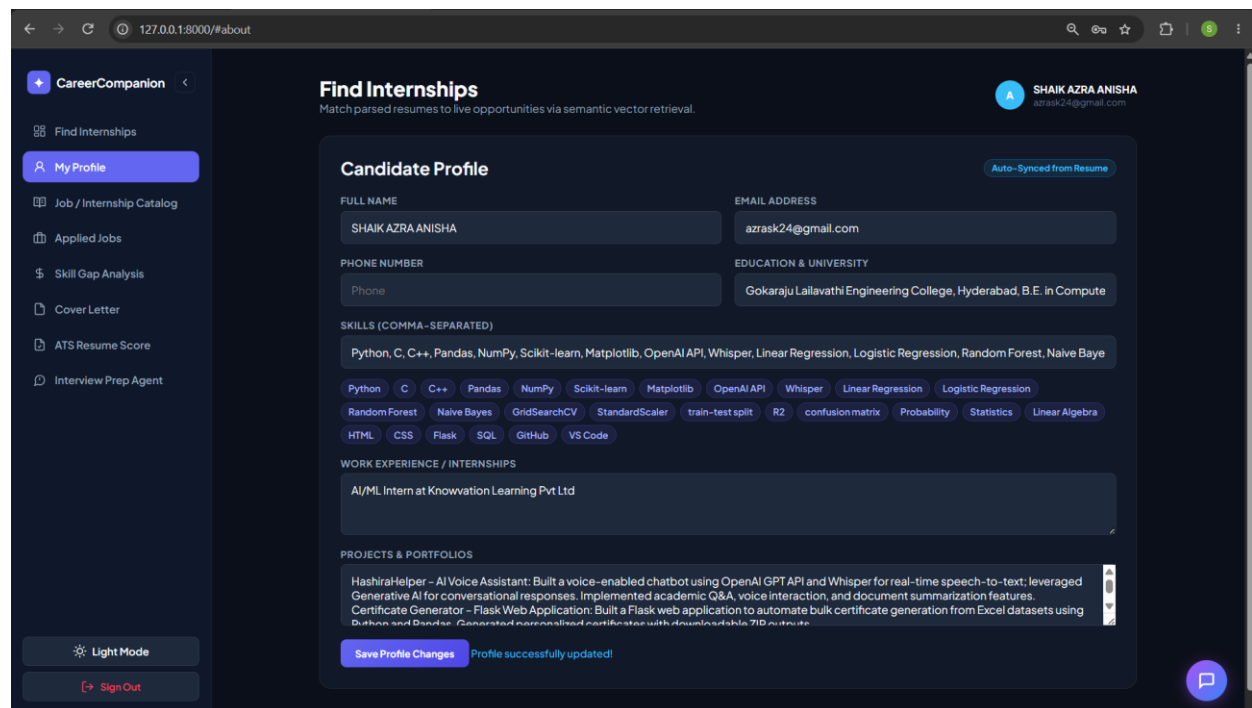
Description:

The User Profile section automatically displays the candidate's information extracted from their uploaded resume.

Implementation:

The structured candidate profile generated during resume parsing is used to populate profile details such as skills, education, and experience.

Output:



Feature 7: Job / Internship Catalog

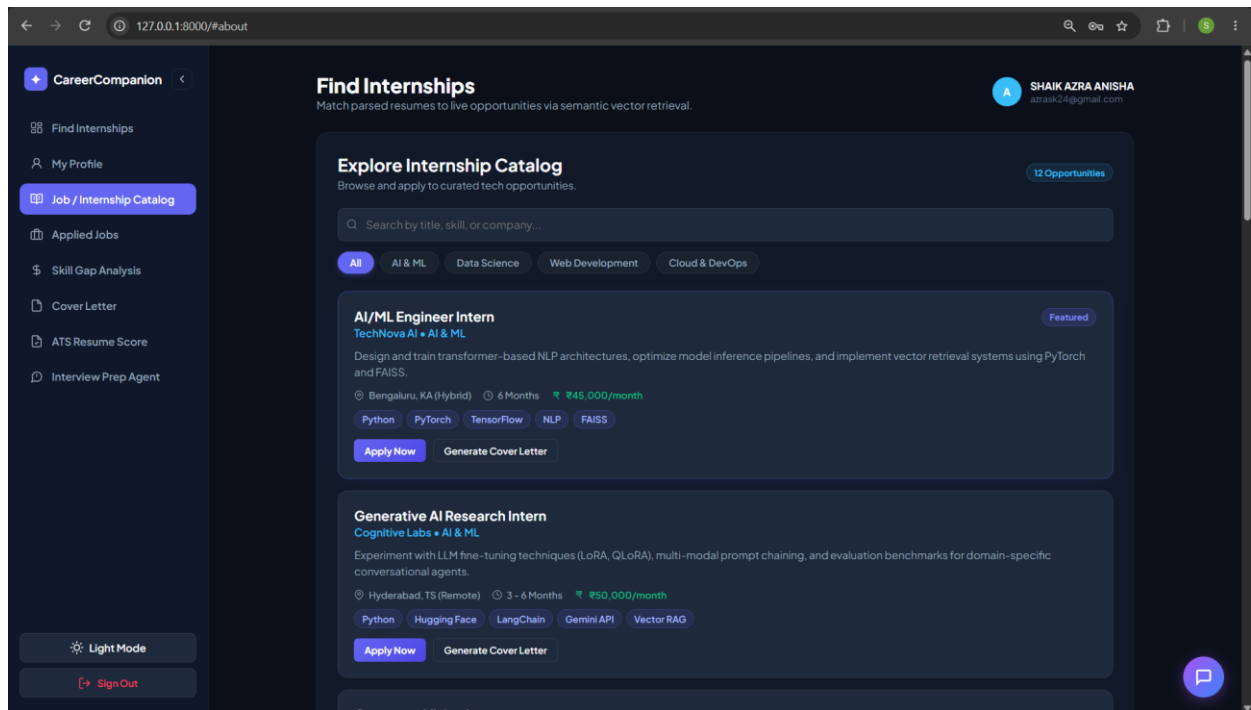
Description:

The platform provides a structured catalog of available internship opportunities for users to explore.

Implementation:

The catalog contains **12 internship roles** with relevant details and INR-based stipend information.

Output:



Feature 8: Application Tracker Applied Jobs

Description:

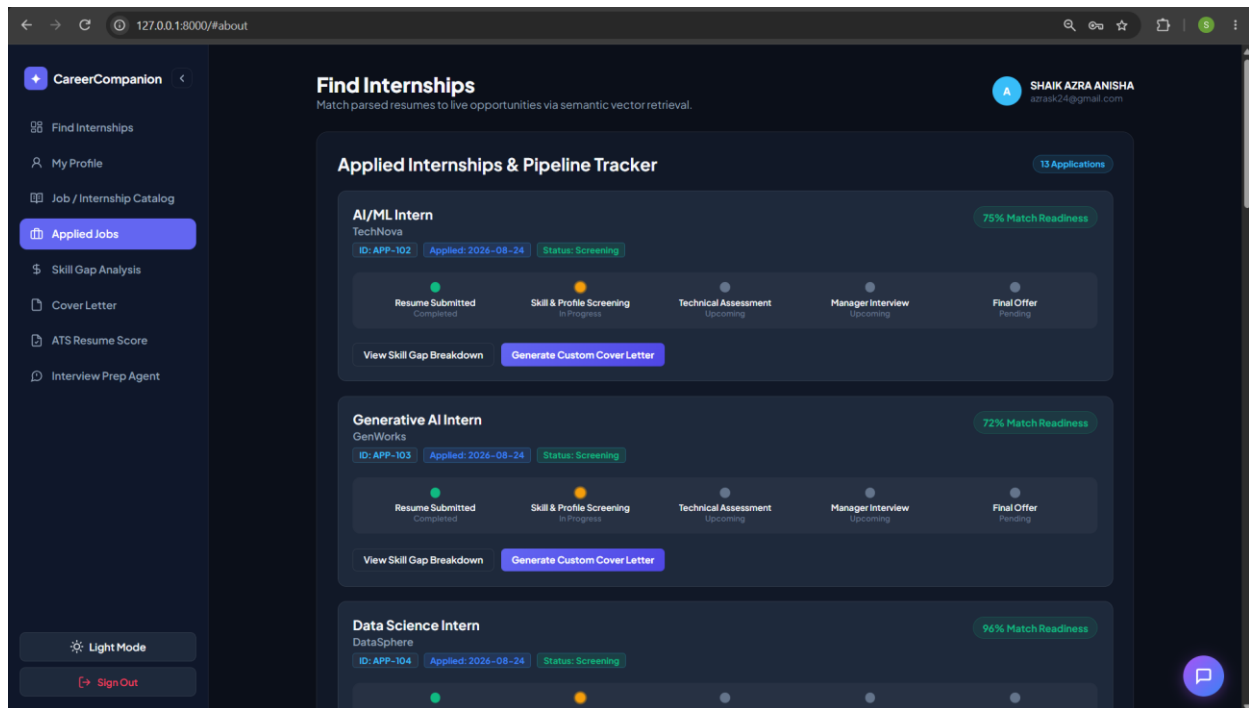
The platform allows users to track internships they have applied for and monitor their application progress.

Implementation:

Applications are tracked through five stages:

Applied → Skill Screening → Technical Assessment → Interview → Offer.

Output:



Feature 9: Skill Gap Analysis

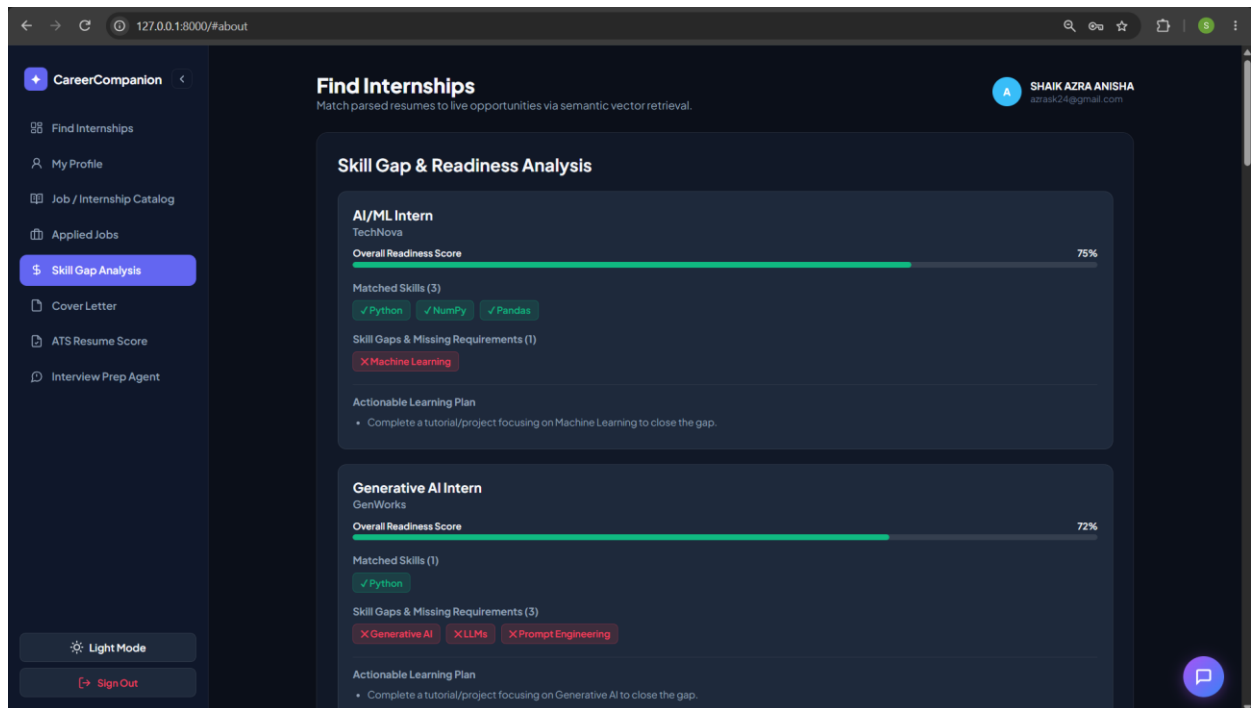
Description:

The system identifies missing or weaker skills by comparing the candidate's profile with internship requirements.

Implementation:

The skill-gap analysis evaluates the candidate's resume information against relevant internship requirements and provides areas for improvement.

Output:



Feature 10: Cover Letter Generator

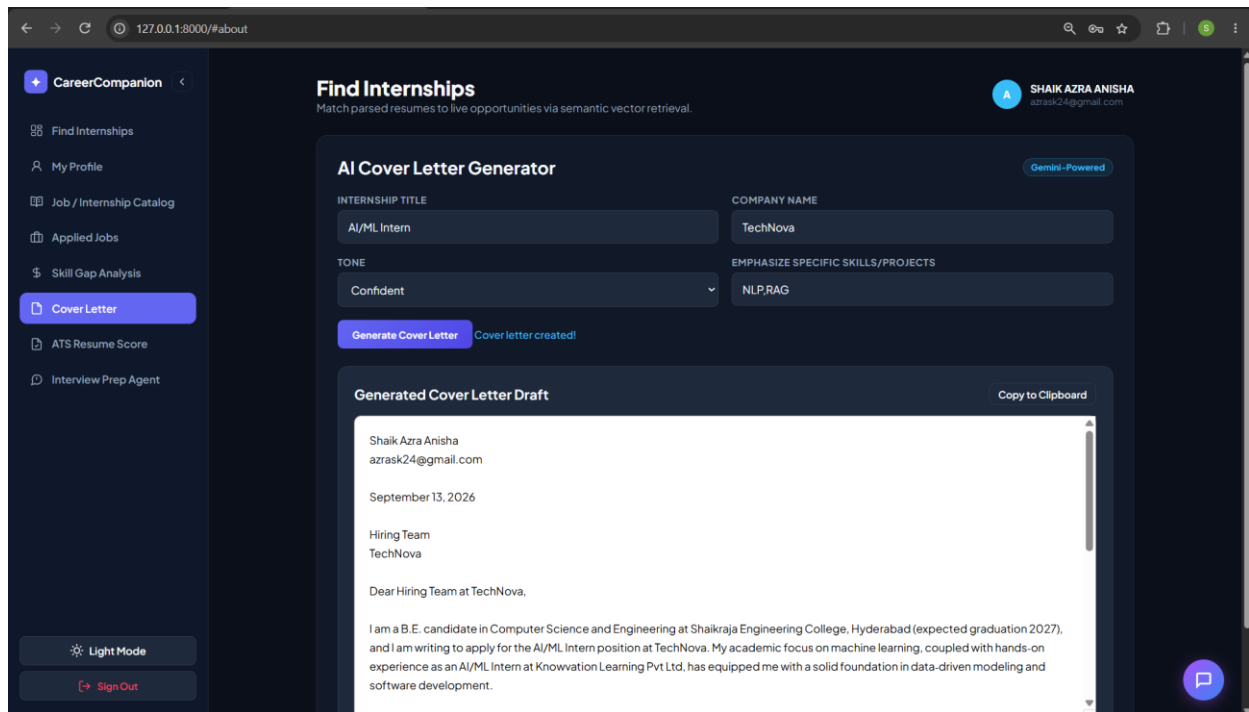
Description:

The system generates personalized cover letters based on the candidate's profile and selected internship.

Implementation:

Candidate information and internship details are provided to the AI generation workflow to create a customized cover letter.

Output:



Feature 11: ATS Resume Score

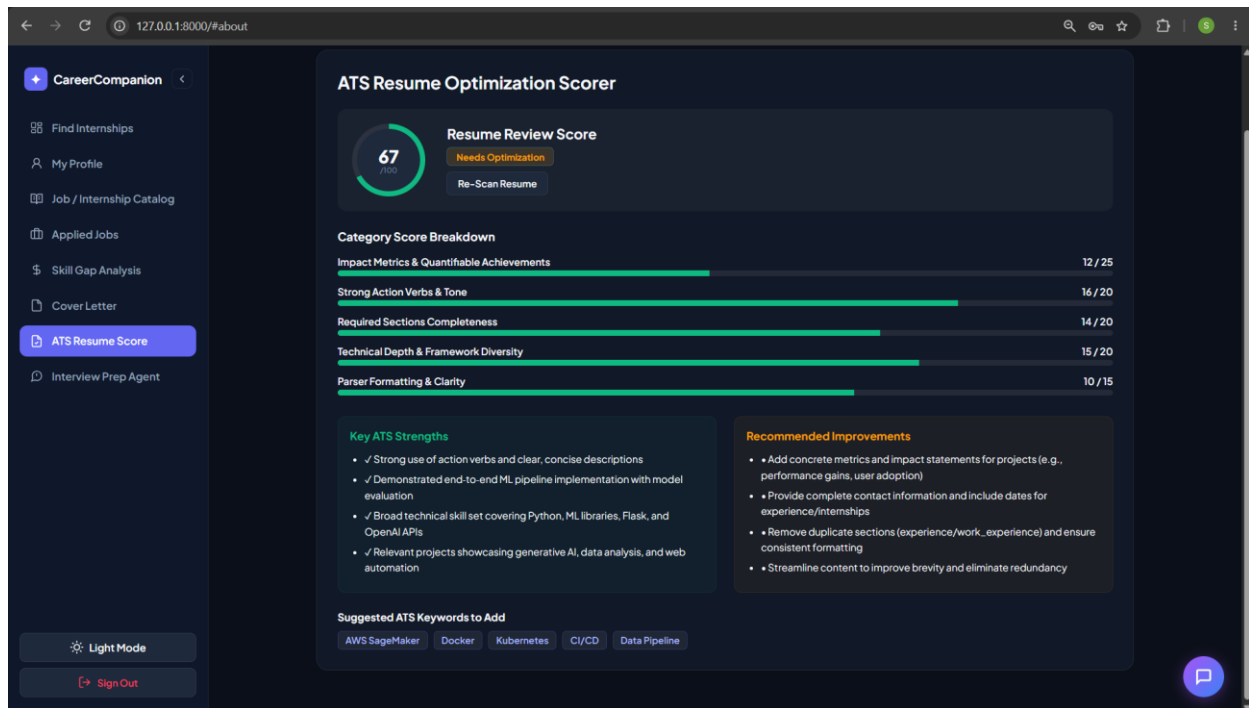
Description:

The system evaluates the resume and provides an ATS score to help candidates improve their resumes.

Implementation:

The ATS analysis evaluates resume information against relevant internship requirements and identifies areas that may affect the candidate's suitability.

Output:



Feature 12: Interview Preparation Agent

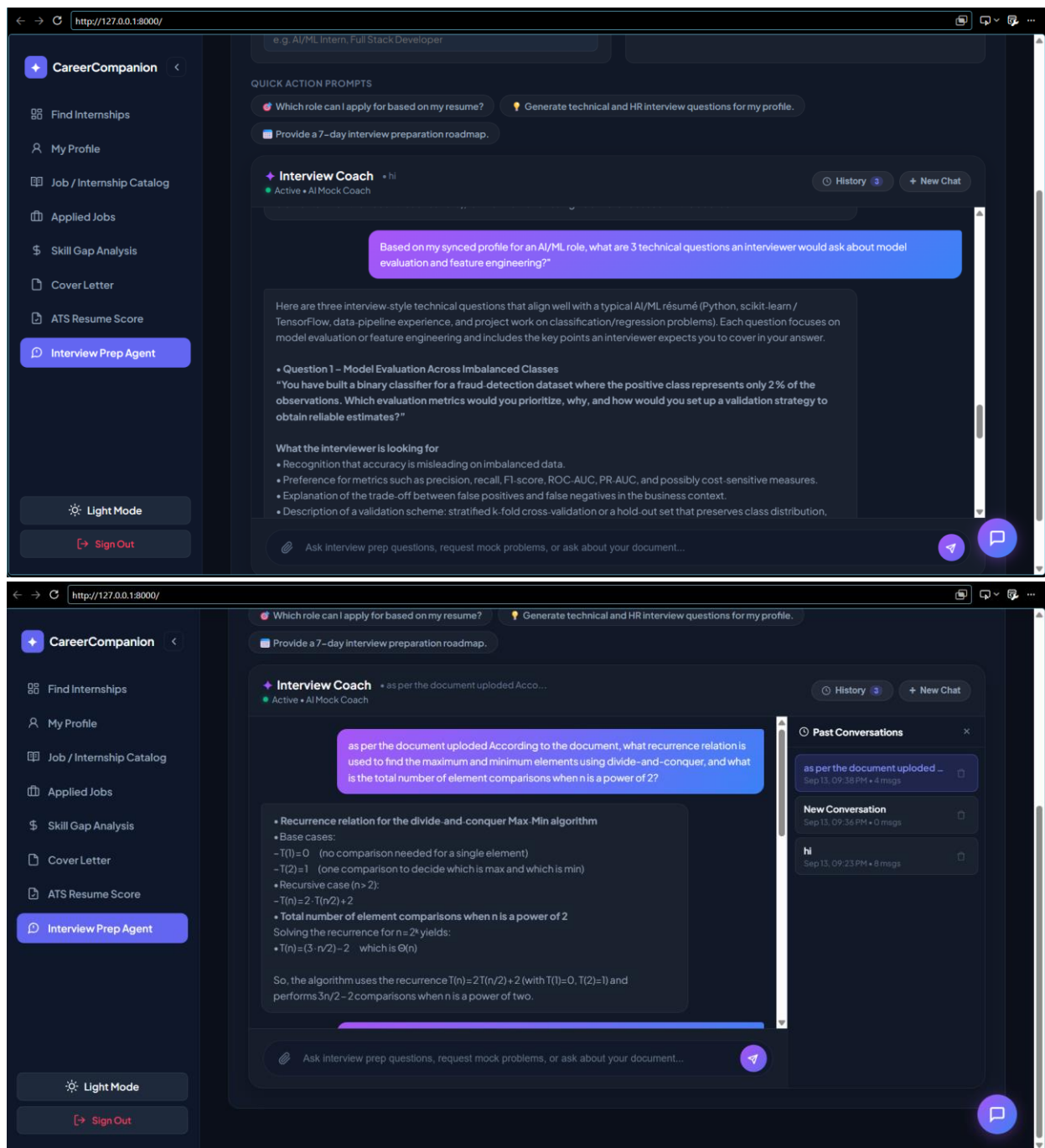
Description:

The Interview Preparation Agent helps candidates prepare for internship interviews through interactive AI-based practice.

Implementation:

The agent uses the candidate's profile and conversation context to generate interview questions and guidance. It follows the **STAR method** for structured behavioral interview preparation and includes anti-cheating guardrails. It also supports **Document-Based Q&A**, allowing candidates to ask questions from uploaded interview preparation documents and receive relevant context-based answers.

Output:



Feature 13: Career Copilot

Description:

The Career Copilot provides interactive AI-based career guidance and assistance.

Implementation:

The Career Copilot uses career-related context to provide guidance and suggestions. Chat sessions use a **chat/session ID** to maintain conversation context during the current session, with policy-based guardrails for relevant responses.

Output:

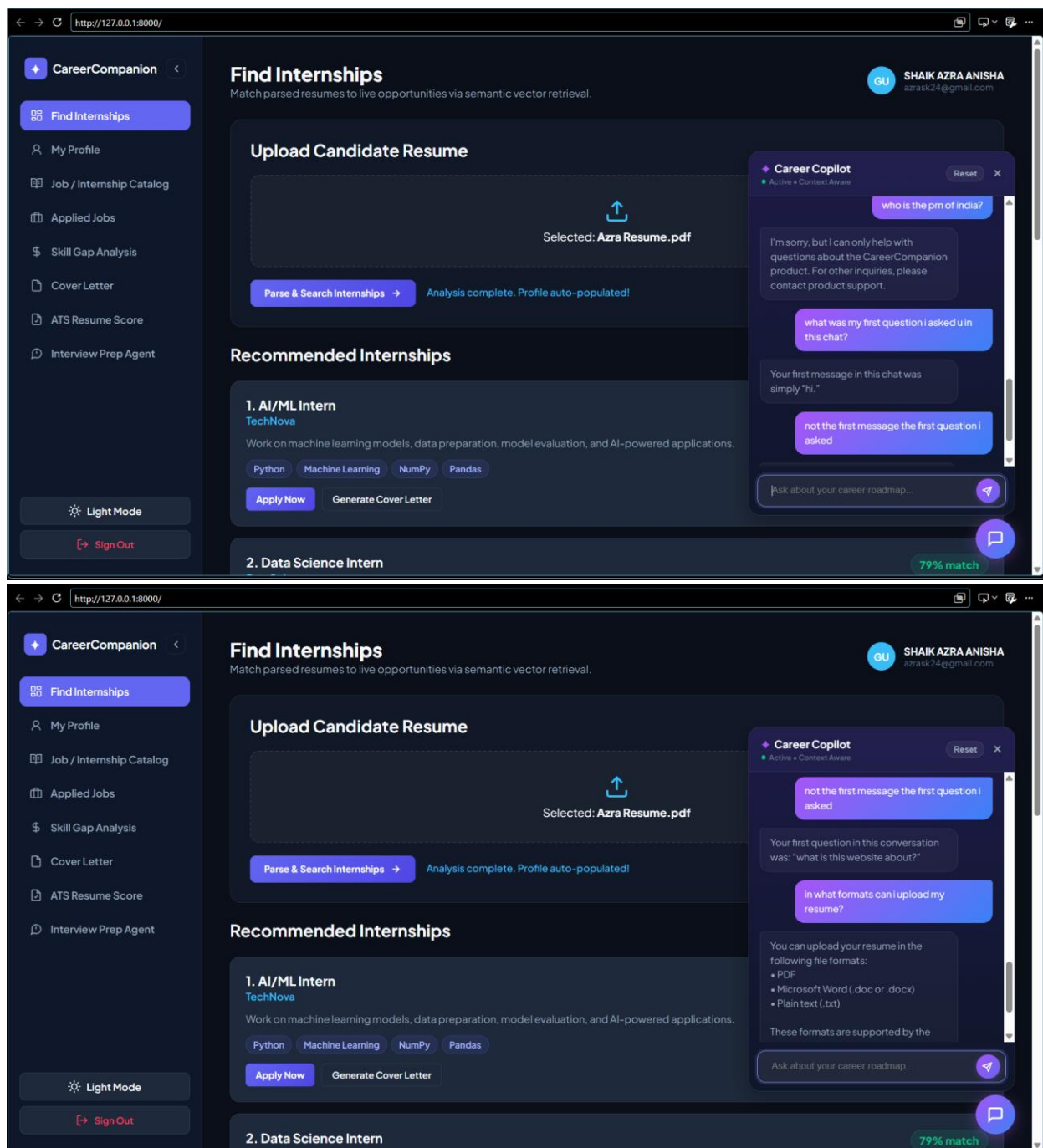
The image displays two screenshots of the CareerCompanion web application, showing the user interface for finding internships. The application is accessed via a browser at the URL `http://127.0.0.1:8000/`.

Top Screenshot:

- Header:** The user is logged in as **SHAIK AZRA ANISHA** (azraak24@gmail.com).
- Left Sidebar:** Contains navigation links: **Find Internships** (active), **My Profile**, **Job / Internship Catalog**, **Applied Jobs**, **Skill Gap Analysis**, **Cover Letter**, **ATS Resume Score**, and **Interview Prep Agent**. At the bottom are **Light Mode** and **Sign Out** buttons.
- Main Content Area:**
 - Find Internships:** Subtitle: "Match parsed resumes to live opportunities via semantic vector retrieval."
 - Upload Candidate Resume:** A dashed box for uploading a resume. Below it, a button **Parse & Search Internships** is shown next to the status **Analysis complete. Profile auto-populated!**. The selected file is **Azra Resume.pdf**.
 - Recommended Internships:**
 - 1. AI/ML Intern TechNova:** Description: "Work on machine learning models, data preparation, model evaluation, and AI-powered applications." Skills: **Python**, **Machine Learning**, **NumPy**, **Pandas**. Buttons: **Apply Now**, **Generate Cover Letter**.
 - 2. Data Science Intern:** A green badge indicates a **79% match**.
- Right Panel (Career Copilot):** A chat interface with a **Reset** button. It shows a greeting: "Hello! How can I help you with CareerCompanion today? You can ask about creating an account, uploading your resume, matching internships, skill-gap analysis, generating cover letters, or tracking applications. If you have any other questions, feel free to let me know." Below this is a user query: **what is this website about?** and a response: "CareerCompanion is an intelligent career-acceleration platform designed for university students and early-career". At the bottom is a prompt: **Ask about your career roadmap...**

Bottom Screenshot:

This screenshot shows the same interface as the top one, but with the **Career Copilot** chat window expanded to show more details. The response to the query "what is this website about?" is now visible, describing the platform as an "intelligent career-acceleration platform designed for university students and early-career software engineers" and listing key features: "Resume parsing - automatically extracts skills, experience, and education from your uploaded resume." and "Semantic matching - uses vector search to recommend internships that closely align with your profile and interests." The **Ask about your career roadmap...** prompt remains at the bottom of the chat window.



Feature 14: Light / Dark Mode

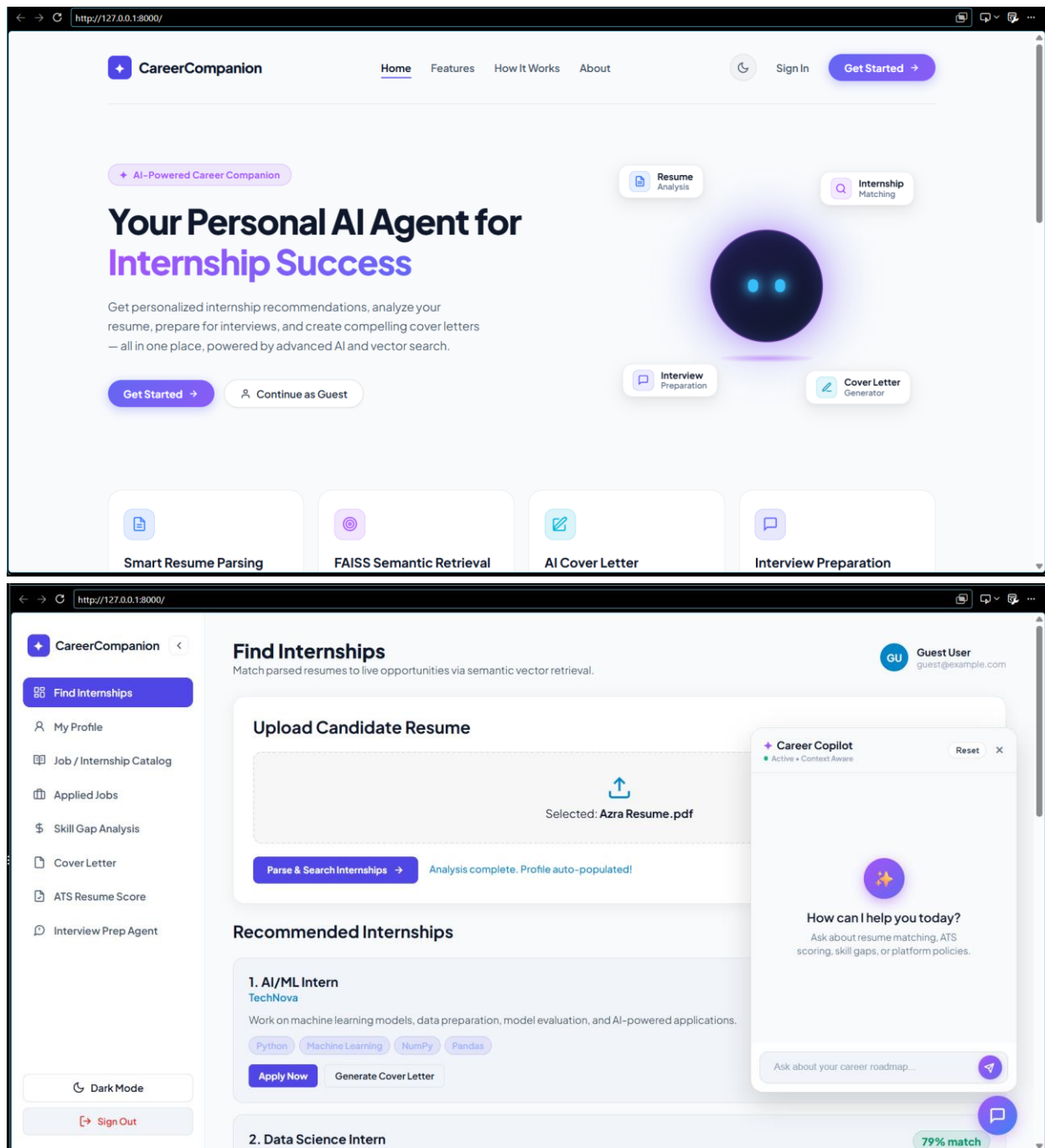
Description:

The platform provides a light and dark mode option that allows users to switch the interface according to their visual preference.

Implementation:

A theme toggle was implemented to dynamically switch the application's interface between Light Mode and Dark Mode. The selected theme is applied consistently across the dashboard, navigation components, cards, and other UI elements.

Output:



10. Testing

Test ID	Test Case	Expected Result	Actual Result	Status
T001	Login with valid credentials	User logs in successfully	User logged in successfully	Pass
T002	Register a new user	User account is created	User account created successfully	Pass
T003	Upload PDF resume	PDF is uploaded and processed	PDF uploaded and processed	Pass
T004	Upload DOCX resume	DOCX is uploaded and processed	DOCX uploaded and processed	Pass
T005	Resume information extraction	Resume details are extracted	Resume details extracted successfully	Pass
T006	Hybrid AI resume parsing	Skills, education, and experience are identified	Information extracted using AI and regex	Pass
T007	Internship matching	Relevant internships are retrieved	Relevant internships displayed	Pass
T008	FAISS semantic search	Suitable internships are retrieved based on similarity	Matching internships retrieved	Pass
T009	ATS analysis	Resume score and improvement areas are generated	ATS analysis displayed	Pass
T010	Skill gap analysis	Missing/relevant skills are identified	Skill gaps displayed	Pass
T011	Cover letter generation	Personalized cover letter is generated	Cover letter generated	Pass
T012	Interview preparation	AI provides relevant interview assistance	Interview preparation response generated	Pass
T013	Invalid file upload	Appropriate error message is displayed	Error message displayed	Pass
T014	End-to-end user workflow	Complete workflow functions correctly	Workflow completed successfully	Pass

11. Challenges and solution

Challenge	Solution
Slow AI responses	Switched from Gemini API to Groq API
Inaccurate resume parsing during initial development	Refined the AI prompt with clearer instructions and structured extraction requirements
Different resume formats	Added PDF/DOCX-specific extraction
Inconsistent resume data	Combined Regex with AI-based parsing
Keyword-based matching limitations	Implemented embeddings and FAISS
Need for relevant AI responses	Implemented RAG-based retrieval
Chatbot unable to retain conversation context	Implemented session/chat ID-based conversation memory
PostgreSQL configuration in an SQLite-based project	Removed the unused PostgreSQL configuration and standardized the project on SQLite
Maintaining application state	Used browser localStorage

12. Final Project Status

Completed Features

- User Authentication & Sessions
- Resume Upload and PDF/DOCX Parsing
- Hybrid Resume Parsing using LLM and Regex
- AI-Based Resume Analysis
- FAISS-Based Internship Matching
- RAG-Based Internship Explanation
- ATS Score & Skill Gap Analysis
- Cover Letter Generator
- Application Tracker
- Interview Preparation Agent
- Document Q&A
- Career Copilot with Conversation Context
- AI Safety & Platform Guardrails
- Internship Catalog
- Dashboard and API Integration
- Collapsible Sidebar & Theme Switching
- Interactive Landing Page Canvas

Testing

- End-to-End API & User Workflow Testing
- Resume Parsing and Internship Matching Testing
- AI Feature Testing
- Responsive UI and Theme Testing

Final Outcome

The project successfully implements an AI-powered career assistance platform that supports resume processing, AI-based resume analysis, semantic internship matching, ATS evaluation, skill-gap analysis, personalized cover letter generation, application tracking, interview preparation, document-based question answering, and interactive career assistance.

The platform integrates a FastAPI backend with a responsive Vanilla JavaScript frontend and uses Groq-based AI processing, FAISS-based semantic search, RAG workflows, and SQLite for application data. The system also includes AI safety guardrails, interview anti-cheating controls, session-based conversation context, and a structured internship catalog.

The major application features have been implemented and integrated into a unified platform. The completed system provides users with an end-to-end workflow for analyzing their resumes, discovering relevant internships, improving their profiles, tracking applications, preparing for interviews, and receiving AI-powered career guidance.